

Dynamics of quasiparticles — Solution

Y23 (2023) — English translation

A1

0.70 pts

Let the dependence of electron energy on momentum have the form

$$\varepsilon(p) = \varepsilon_0(3 - \cos(ap_x) - \cos(ap_y) - \cos(ap_z)).$$

Find *его* acceleration at the moment времени, когда momentum equals $p_x = \frac{\pi}{2a}$, $p_y = \frac{\pi}{3a}$, $p_z = 0$, а компоненты действующей на electron силы F_x , F_y , F_z .

Let's express the speed of the electron in terms of its momentum:

$$v_x = \frac{\partial \varepsilon}{\partial p_x} = \varepsilon_0 a \sin(ap_x),$$

similarly for остальных двух компонент

$$v_y = \varepsilon_0 a \sin(ap_y), \quad v_z = \varepsilon_0 a \sin(ap_z).$$

From the equation движения $\frac{d}{dt}\vec{p} = \vec{F}$. Let us express acceleration via/through производную momenta:

$$a_x = \dot{v}_x = \varepsilon_0 a^2 \cos(ap_x) \dot{p}_x = \varepsilon_0 a^2 \cos(ap_x) F_x.$$

The formulas for the remaining components are similar. For impulse values from the condition

Answer: \textbf{Answer:}

$$a_x = 0, \quad a_y = \frac{1}{2}\varepsilon_0 a^2 F_y, \quad a_z = \varepsilon_0 a^2 F_z.$$

A2

0.80 pts

Let an electron with $\varepsilon(\vec{p})$ from A1 move in a constant electric field of magnitude E directed along the x axis. The initial coordinates and projections of the momentum are equal to zero. Electron charge $-q$. Find his law of motion $x(t)$.

The electron is acted upon by the force $F_x = -qE$, so the momentum

$$p_x = -qEt,$$

а соответствующая velocity

$$v_x = -a\varepsilon_0 \sin(aqEt).$$

Остальные проекции скорости equal нулю. Integrating, let us find зависимость координаты x as a function of time

Answer: \textbf{Answer:}

$$x(t) = -\frac{\varepsilon_0}{qE}(1 - \cos(aqEt))$$